

Read the following Instructions.

1. Read the scenario carefully before answering.
 2. Provide clear explanations with proper reasoning.
 3. Avoid copying sentences directly from the scenario.
 4. Use diagrams where necessary.
 5. Justify your answers using proper principles. Clearly mention any assumptions made.
-

Scenario

You are part of a software engineering team hired to analyse and design an online food delivery platform similar to modern delivery applications. The goal is to create a system that connects customers, restaurants, delivery personnel, and administrators through a single digital platform. The system must support:

- ★ User registration and login
- ★ Restaurant browsing and menu viewing
- ★ Cart management and order placement
- ★ Secure online and cash payment options
- ★ Real-time order tracking
- ★ Restaurant profile and menu management
- ★ Order processing and status updates
- ★ Delivery assignment and route management
- ★ Administrative monitoring and analytics

The company plans to expand the platform in the future by:

- ★ Supporting thousands of simultaneous users
- ★ Adding multiple cities and regions
- ★ Integrating AI-based recommendations
- ★ Introducing loyalty programs and promotional offers
- ★ Supporting third-party payment gateways and delivery partners

As an analyst, your task is to study the scenario carefully and produce analytical and design-based answers.

Questions

01. Identify and describe at least FIVE high-level requirements of the online food delivery system. For each requirement:

- I. Provide a suitable requirement name.
- II. Explain the requirement clearly.
- III. Identify the stakeholders involved.
- IV. Explain why it is considered a high-level abstraction rather than a low-level implementation detail.
- V. Mention one possible challenge in implementing that requirement.

Example Structure

Requirement	Description	Stakeholders	Why it is a High-Level	Challenge
User Authentication	Allows to securely register and log in	Customers, Admins	Focuses on system capability	Protecting user credentials

02. Explain why abstraction is useful during early system design.

03. Identify and explain at least SIX major modules of the system. (Use a table format to answer this)

- I. Provide the module name.
- II. Describe its primary responsibilities.
- III. Identify the data it handles.
- IV. Explain which other modules it interacts with.

04. After identifying the modules, answer the following

- I. Why is decomposition important when developing large-scale systems?
- II. How does modular design support teamwork during software development?
- III. What problems might occur if the system is developed as one large component instead of separate modules?
- IV. Which module do you think is the most critical in this system? Justify your answer.
- V. Which module is likely to experience the highest system load? Explain why.

05. Design a detailed flowchart for the process of placing an order in the online food delivery system. Your flowchart MUST include:

User login, Restaurant browsing, Menu viewing, Item selection, Cart review, Order confirmation, Decision point for payment method, Cash payment process, Card payment process, Order cancellation option before final confirmation, Restaurant notification, Order preparation, Delivery assignment, Delivery tracking, Order completion

Mandatory Decision Points Your flowchart should clearly show:

- What happens if login fails?
 - What happens if the user cancels the order?
 - What happens if card payment fails?
 - What happens if a restaurant rejects the order?
 - What happens if delivery personnel are unavailable?
-

Submission Guidelines

- This is a Individual work and submission
- Use clear headings for each section
- Diagrams may be hand-drawn or software-generated
- Flowcharts must use standard symbols
- Maintain proper formatting and readability for your answers
- Plagiarisms will be checked during the evaluations
- Zip and Rename your files before the submission : **LAB01_INDEXNUMBER**